

Sri Lanka Institute of Information Technology

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Lab Exercise 01



Details

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Activity 1 - Stakeholder Identification

Stakeholder Categorization for Fit Flow

Stakeholder Category	Stakeholder	Why they belong in this category
Primary Stakeholders	Current FitFlow users	People currently using FitFlow daily/weekly for workouts and tracking.
	Lapsed users	Former users who stopped using the app. (the group driving the retention drop)
	Fitness beginners	New users like Priya Singh's persona, needing guidance and easy onboarding.
	Busy professionals	Users like Alex Rivera who need quick, adaptive workouts around irregular schedules.
	Premium/paying subscribers	Users who pay for advanced features and expect reliability.
Secondary Stakeholders	Family members	May use or be affected by the app's social/sharing features
	Personal trainers / gym staff	Recommend the app to clients or work alongside it during sessions
	Healthcare professionals	Interested in anonymized wellness data and may refer patients to fitness apps
	Workout/accountability partners	Friends or peers connected through the app's social community feature
	Gym partner organizations	Businesses collaborating with FitFlow, indirectly shaping user experience
Tertiary Stakeholders	Product Manager (Priya Sharma) and design team	Own the product direction and decisions from research findings

	Engineering and data science team	Build the AI workout engine and nutrition computer-vision features
	CEO and business leadership	Focused on revenue recovery and overall company performance
	Marketing team	Responsible for the phased launch and messaging around the redesign
	Regulatory bodies	Oversee data privacy compliance (GDPR/CCPA) for the app

Importance of Each Stakeholder Group

- **Primary stakeholders** are important because they are the actual people experiencing the pain points behind Fit Flow's declining ratings and retention. Understanding their frustrations directly shapes what the redesign needs to fix.
- **Secondary stakeholders** matter because they influence whether primary users adopt or stay engaged with the app for example, a personal trainer's recommendation or a family member joining a challenge can affect a user's motivation to keep using Fit Flow.
- **Tertiary stakeholders** are important because they control the resources, technical feasibility, and business/legal direction of the project. Without their involvement, insights from user research can't be turned into an actual product.

Activity 2 – Selection of Research Methods

Selected methods:

Semi-structured interviews, surveys, and contextual inquiry (observation).

Justification:

For the Fit Flow redesign, a mix of interviews, surveys, and contextual inquiry gives the most complete picture of user behavior without relying on just one data source. Surveys reach a large number of both active and lapsed users quickly, making it possible to measure things like satisfaction levels, usage frequency, and interest in features such as AI workouts or nutrition tracking across a broad group. Semi structured interviews go deeper with a smaller set of users,

uncovering the reasons behind the drop in ratings such as frustration with generic workout plans or difficulty logging meals that a survey alone wouldn't reveal. Contextual inquiry, where users are observed working out or logging food in their normal environment, shows real usage patterns and friction points that people often don't mention or even notice when simply asked in an interview or survey. Together, these three methods complement each other: the survey identifies what is happening on a scale, the interviews explain why it's happening, and the observation confirms how it happens in practice.

Combining qualitative and quantitative approaches:

The survey will provide the quantitative data closed and scale-based questions producing numbers that can be compared across user groups, such as percentage of users who find nutrition logging tedious or average satisfaction scores. The interviews and observations will provide the qualitative data direct quotes, described feelings, and observed behaviors that explain the reasoning behind those numbers. These will be analyzed together during the findings stage, using the survey statistics to show the scale of a problem and the interview/observation insights to explain the cause, giving both breadth (how many people are affected) and depth (why they are affected) to the research.

Activity 3 – Research Instruments

1.Semi-Structured Interview Guide

Thank you for participating. This interview is part of research to understand your experience with Fit Flow and improve future features. There are no right or wrong answers, and you can skip any question or stop at any time.

1. How long have you been using Fit Flow, and how often do you use it in a typical week?
2. Can you walk me through what you usually do when you open the app?
3. What was your first impression of Fit Flow when you started using it?
4. What do you find most useful or enjoyable about the app?
5. Can you describe a time when you felt frustrated or confused while using Fit Flow?
6. How do you currently plan or choose your workouts within the app?
7. What is your experience like when logging for meals or tracking nutrition?
8. Do you interact with other users through the app? If so, how, and how do you feel about it?
9. Have you ever stopped using the app for a period? If yes, what led to that?
10. How would personalize AI workout recommendations improve your fitness experience?
11. What would make you feel more motivated or connected while using a fitness app like this?
12. Is there anything about your current fitness routine that Fit Flow doesn't support well?

2. Survey Questionnaire

Section 1: Background

1. How long have you been using Fit Flow? (Multiple choice: Less than 1 month / 1-6 months / 6-12 months / Over 1 year)
2. How often do you use the app? (Multiple choice: Daily / A few times a week / Weekly / Rarely)

Section 2: Current Experience (Scale questions, 1 = Strongly Disagree, 5 = Strongly Agree)

3. Fit Flow's workout plans feel personalized to my needs.
4. Logging up my meals and nutrition is easy and quick.
5. I feel motivated to keep using Fit Flow.
6. I feel connected to other users through the app.
7. Fit Flow's app performance (speed, reliability) meets my expectations.

Section 3: Pain Points

8. What is the most frustrating part of using FitFlow? (Open-ended)
9. Have you ever considered stopping or actually stopped using the app? (Multiple choice: Yes, currently not using / Yes, but came back / No)
10. If yes, what was the main reason? (Open-ended)

Section 4: New Features

11. How interested are you in AI-generated personalized workout plans? (Scale: 1-5)
12. How interested are you in social features like group challenges or friend connections? (Scale: 1-5)
13. How interested are you in improved nutrition tracking (e.g., camera-based food logging)? (Scale: 1-5)
14. Which new feature would you be most excited to try first? (Multiple choice: AI workouts / Social community / Nutrition tracking / None of these)

Section 5: Final Thoughts

15. Is there anything else you'd like FitFlow to improve or add? (Open-ended)

3. Observation / Contextual Inquiry Checklist

- ☐ Participant consent confirmed before observation begins
- ☐ Device/environment used noted (phone model, home/gym setting)
- ☐ Time taken to complete onboarding or start a workout
- ☐ Number of taps/screens needed to log a meal
- ☐ Any hesitation, confusion, or repeated actions observed
- ☐ Verbal comments or reactions made while using the app (note down directly)
- ☐ Whether participant refers to help/support features
- ☐ Interaction with any social or community features, if applicable
- ☐ Participant's immediate reaction after completing the task
- ☐ Any workaround behavior observed (e.g., using a separate app for nutrition tracking)
- ☐ Researcher's overall notes on friction points and positive moments

Activity 4 – Pilot Testing

Pilot Testing Process

Before conducting the main user research, the interview guide, survey questionnaire, and observation checklist were pilot tested with three classmates. The purpose of the pilot test was to evaluate whether the questions were clear, easy to understand, and relevant to the Fit Flow redesign project. Participants were also asked to comment on the overall length, wording, and any confusing or repetitive questions.

The pilot sessions were conducted individually and took approximately 15–20 minutes per participant. Each participant first completed the survey, followed by a short interview. During the session, observations were recorded using the observation checklist. After completing the activities, participants provided feedback on the research instruments.

Pilot Test Feedback

Participant	Clarity	Length	Difficulty	Feedback Received
Classmate 1	Good	Appropriate	Easy	Suggested simplifying Question 10 in the interview because “AI-generated personalized workout plans” was too technical.
Classmate 2	Very Good	Slightly Long	Easy	Felt the survey flowed well but recommended moving all demographic questions to the beginning for better organization.
Classmate 3	Good	Appropriate	Moderate	Suggested adding a short explanation before the observation session so participants knew what tasks they were expected to perform.

Changes Made After Pilot Testing

Issue Identified	Improvement Made
Technical wording in Interview Question 10.	Reworded it to “How would personalized AI workout recommendations improve your fitness experience?”
Survey structure could be improved.	Grouped demographic questions at the beginning, followed by experience, pain points, and feature-related questions.
Participants were unsure about observation tasks.	Added a short briefing explaining the purpose of the observation and the tasks to be completed before the session started.
Some open-ended questions produced very short answers.	Added follow-up prompts during interviews were appropriate to encourage participants to provide more detailed responses.

Summary of Pilot Testing

The pilot testing successfully identified several small improvements that increased the clarity and usability of the research instruments. None of the participants experienced major difficulties completing the interview or survey, indicating that the overall design of the instruments was appropriate for the Fit Flow research. The feedback helped improve the wording of certain questions, organize the survey more logically, and provide clearer instructions for observation sessions. As a result, the final research instruments were considered suitable for collecting reliable and meaningful data during the main user research phase.

Activity 5 – User Research Findings

Participants: 5 total, representing 2 stakeholder groups — Regular Users (3) and Fitness Trainers (2), consistent with the primary/secondary stakeholders identified in Activity 1.

Participant	Stakeholder Group	Age	Usage
P1	Regular User	25	Active, uses daily
P2	Regular User	32	Lapsed, stopped 2 months ago
P3	Regular User	21	Active, uses a few times/week
P4	Fitness Trainer	34	Uses app to guide clients
P5	Fitness Trainer	30	Occasional user, mostly reviews client data

Session Notes & Raw Findings

P1 — Regular User (Interview + Observation)

Uses FitFlow daily for logging workouts. Said the workout plans “feel repetitive after a few weeks” and wished the app adjusted based on progress. During observation, took noticeably longer navigating the nutrition logging screen, tapping back and forth twice before finding the “add meal” button. Reacted positively to the idea of AI-personalized plans: “If it actually learned what I like, I'd probably stick with it longer.”

P2 — Regular User (Interview)

Stopped using the app two months ago. Main reason given: “Logging food every day got tedious, and the plans didn't feel like they were made for me.” Mentioned feeling no sense of progress or reward, calling the app “just a checklist.” Expressed interest in social/community features if they didn't feel forced.

P3 — Regular User (Survey + Interview)

Currently active, uses app 4–5 times weekly. Survey responses rated personalization 2/5 and motivation 3/5. In interview, said the app “does the job but doesn't feel exciting,” and would like small celebratory moments (e.g., streaks, badges) for consistency. Showed strong interest in social challenge features.

P4 — Fitness Trainer (Interview + Observation)

Uses FitFlow to monitor a few clients remotely. Said the biggest gap is the lack of a way to review a client's progress or adjust their plan directly. During observation, struggled to locate any client-facing progress summary, spending time searching through multiple tabs. Suggested a dedicated trainer/coach view.

P5 — Fitness Trainer (Survey + Interview)

Uses the app occasionally, mainly to understand what clients see. Rated the AI workout concept highly (5/5 interest) but was skeptical about accuracy: “It depends on whether the AI actually

understands proper form and progression, not just repeats generic routines.” Also noted nutrition tracking felt “built for a general audience, not real training goals.”

Key Findings Summary

Theme	Supporting Evidence
Personalization gap	P1, P2, P3, P5 all noted workout/nutrition plans feel generic
Nutrition logging friction	P1 (observed delay), P2 (main reason for lapsing)
Motivation/reward gap	P2 and P3 both wanted more sense of progress or celebration
Missing trainer tools	P4 and P5 both highlighted lack of a coach/trainer-facing view
Interest in AI features	Positive across users, but trainers (P5) want accuracy/trust, not just novelty

Activity 6 – Consent Form and Data Management

1. Participant Consent Form

FitFlow User Research – Participant Consent Form

- **Study Purpose:**

This research is being conducted as part of a Human-Computer Interaction project to understand user experiences with the FitFlow fitness app, identify pain points, and gather feedback on potential new features (AI workout plans, social community features, and nutrition tracking).

- **Voluntary Participation:**

Participation in this study is entirely voluntary. You may choose not to answer any question and may withdraw from the study at any time, without providing a reason and without any negative consequence.

- **Data Usage:**

Information collected during this session (interview responses, survey answers, and observation notes) will be used only for academic research purposes related to this HCI coursework. Data will not be shared with any third party or used for commercial purposes.

- **Recording Permission:**

With your permission, this session may be audio recorded / notes may be taken to ensure accuracy during analysis. Please indicate below whether you consent to this.

☐ I agree to audio recording

☐ I do not agree to audio recording (notes only)

- **Right to Withdraw:**

You may withdraw from this study at any point during or after the session. If you withdraw, any data collected from you will be deleted and not used in the final research report.

- **Confidentiality:**

Your identity will remain confidential. Your name will not be used in any report or presentation; instead, a participant code (e.g., P1, P2) will be used to refer to your responses.

- **Consent:**

I have read and understood the information above. I voluntarily agree to participate in this research study.

Participant Name: _____

Signature: _____

Date: _____

2. Raw Data Storage and Organization

Raw data collected from the 5 participants (interview responses, survey answers, and observation notes) will be organized and stored as follows:

- Interview notes and recordings will be saved as individual files labeled by participant code (e.g., P1_Interview.docx, P1_Audio.mp3), stored in a dedicated project folder separated from personally identifiable information.
- Survey responses will be exported and stored in a single spreadsheet, with each row representing one participant's responses, again identified only by participant code.
- Observation checklists will be scanned/saved digitally per participant session and stored alongside the corresponding interview file.
- All raw data files will be kept in a single secure folder (e.g., a password-protected local drive or restricted-access cloud storage such as Google Drive with limited sharing permissions) accessible only to the research team members involved in this project.

3. Data Anonymization and Security Measures

- Participant names are replaced with codes (P1–P5) in all notes, transcripts, and summary reports; the key linking codes to real names is stored separately from the research data itself.
- Any identifying details mentioned during interviews (e.g., workplace, specific location) are removed or generalized in the written notes before analysis.
- Signed consent forms are stored separately from the anonymized research data, so that even if research notes were accessed, they could not be linked back to a specific individual without the separate consent records.
- Digital files are stored on password-protected devices/accounts, and access is limited to the research team only.
- Once the research project and report are complete, raw data will be retained only as long as required for coursework evaluation and then securely deleted.